

PROFESSIONAL SUMMARY

Environmental Engineer and ESG Technical Intern with hands-on experience building GHG inventories (Scope 1–3) and life cycle assessments aligned with the GHG Protocol and ISO 14064. Skilled in translating emissions data into audit-ready disclosures (SEBI BRSR) and environmental risk assessments for industrial and energy-adjacent sectors. Seeking to apply carbon accounting and reporting expertise to product carbon footprinting and Scope 3 estimation in a global energy context.

CORE SKILLS

GHG & Carbon Accounting: GHG Protocol (Scope 1, 2, 3), ISO 14064 (Principles), Life Cycle Assessment (LCA), Product Carbon Footprinting exposure, Corporate Carbon Inventories, Emission Factor Application (CEA India, IPCC, DEFRA)

ESG & Sustainability Reporting: SEBI BRSR & BRSR Core, GRI Standards, CSRD (working knowledge), Double Materiality, ESG KPI Mapping & Gap Analysis, ESG Assurance Fundamentals

Data & Analytics: Advanced Excel (VLOOKUP, Pivot Tables, Multi-Sheet Models), SQL, Power BI, Python, QGIS, OpenLCA

Environmental Engineering: Wastewater Treatment Systems, Energy Audits, ZLD/FGD Systems, Environmental Risk Assessment, Phytoremediation

PROFESSIONAL EXPERIENCE

EnviroWealth Services Pvt. Ltd. — Remote

ESG Technical Intern | Nov 2025 – Mar 2026

- Developed and maintained **Scope 1, 2, and 3 GHG emissions estimation methodologies**, building AI-enabled, audit-ready inventories aligned with the GHG Protocol and ISO 14064 principles.
- Designed a value chain data collection template capturing supplier GHG emissions, water usage, waste, and governance indicators for BRSR Principle 6 compliance.
- Created a 5-tab BRSR Core reporting template calculating GHG emissions, energy consumption (GJ), water withdrawal/discharge, waste metrics, and safety indicators (LTIFR) aligned with SEBI mandatory disclosure requirements.
- Compiled an emission factor reference database from CEA CO2 Baseline 2024, India GHG Program, IPCC 2006, and DEFRA 2023 for fuels, transport, and waste streams.

Tamil Nadu Water Supply and Drainage Board (TWAD) — 16 MLD Facility

Environmental Engineering Intern | Jun 2025 – Jul 2025

- Conducted process, hydraulic, and energy audits for a 16 MLD wastewater treatment plant; analyzed operational and electromechanical data to identify efficiency gaps.
- Developed recommendations resulting in an estimated 7% improvement in energy efficiency, with direct relevance to Scope 2 emissions reduction.

PROJECTS & RESEARCH

Environmental Risk & ESG Analysis of NLCIL — M.E. Dissertation

- Applied the Source–Pathway–Receptor (SPR) framework to assess heavy metal contamination risks in the Neyveli industrial region, evaluating environmental and social risk pathways associated with **coal-based industrial/energy operations** — directly transferable to product carbon footprinting in energy assets.
- Performed BRSR-aligned ESG KPI mapping and gap analysis, benchmarking findings to GRI 12 (Coal Sector) and bridging academic research with corporate reporting frameworks.

Tools & Automation

- Built an automated data acquisition tool that ingests data from everyday apps into a centralized backend, streamlining reporting workflows and eliminating manual data entry. (github.com/arunachalamvenkatachalapathy-dev)

ACHIEVEMENTS

- Selected Post-Graduate Researcher, Re-Imagining Urban Rivers (Season 6) — National Institute of Urban Affairs (NIUA) & River Cities Alliance, Government of India.
- Awarded a Rs. 50,000 competitive research grant with national-level mentorship support (Jal Sakthi Ministry funded).

EDUCATION

M.E. Environmental Engineering — Alagappa Chettiar Govt. College of Engineering & Technology, 2026 | CGPA: 9.0/10
Karaikudi

B.E. Civil Engineering — Government College of Engineering, Tirunelveli 2022 | CGPA: 7.8/10

CERTIFICATIONS

Sustainable Finance — UN CC:e-Learn • CSRD Fundamentals • GIS for Climate Change — Esri